

Performance Nutrition

Life is subtle—only through improving on the little things will you succeed in the big things.

The foods and beverages you consume play a primary role in determining your mental acuity, physical energy, and ability to recover from vigorous training or climbing. Therefore, a thoughtfully designed diet will provide a noticeable edge in performance, whereas engaging in a “see-food diet”—you see food and you eat it!—will continue to hamper your performance in a covert way that you may never recognize.

While it’s impossible to say exactly how big a part diet plays in climbing performance, I estimate that most climbers can realize a 10 to 20 percent improvement in their training, recovery, concentration, energy, and overall climbing performance if they dedicate themselves to improved dietary surveillance.

As a dedicated athlete and performance coach, I follow the changing trends in nutrition with great interest. In the late 1980s high-carbohydrate, low-fat diets were the rage; in the mid-1990s high-protein diets were in vogue; in the 2000s the high-fat, low-carbohydrate diets sold the most books and got the greatest hype; and now in the 2010s the Paleo diet has become somewhat popular. Interestingly, all these diets seem to have some scientific backing, yet it’s often hard to separate science from dogma. Fortunately, performance nutrition is not that complex a subject—if you make sensible food choices, eat moderately, and avoid excessive amounts of sugar, alcohol, and unhealthy fats, then you’ll be eating more healthfully than most folks.

I was fortunate to realize the causal connection between nutrition and performance many years ago. Though I occasionally relax my dietary surveillance, I credit sound nutrition—about forty-eight weeks per year—for my ability to train exceedingly hard, avoid injury, recover quickly, and still climb at a high level into my fifties—despite decades of climbing abuse to my joints, tendons, and muscles. Follow the guidelines presented in this chapter, and I trust that you can do the same as well!

Macronutrients

There is no single perfect diet for climbers, just as there is no single perfect training program. To some extent the amount and best type of foods for you depend on your climbing preference. For instance, alpine climbers have significantly different nutritional needs and energy requirements than folks who partake in cragging or bouldering.

Our study of performance nutrition begins with a look at the three macronutrients: protein, fat, and carbohydrate. As in the prior chapters, the focus will remain on the best strategies for rock climbers, with only general information for alpine climbers.

Protein

Protein has many functions in the body, including building and repairing tissue, acting as a major component of the immune system, and making up enzymes, which facilitate every reaction that goes on in the body. Growing individuals need more protein than adults do simply because they are actually laying down large amounts of new tissue.

Healthy adults have a fairly extensive protein pool to draw on—that is, the proteins we consume are recycled several times for different functions in the body. For this reason daily protein requirements for adults are modest, even if they are training to increase muscle mass. Successful training is much less a factor of consuming a lot of protein than of using the appropriate training strategy and eating enough carbohydrates to fuel your training.

DAILY REQUIREMENTS

Between 1.2 and 1.5 grams of protein per kilogram of body weight per day is adequate for most climbers. For a seventy-two-kilogram (160-pound) individual, this translates to 86 to 108 grams per day. This is higher than the 0.8 to 1.0 gram per kilogram recommended for sedentary individuals by the FDA. Some studies have shown a slightly higher need in athletes, not just to increase muscle mass but also to facilitate recovery from exercise and compensate for the catabolic (tissue-consuming) effects of long, intense exercise. Still, I (and most nutritionists) do not buy in to the massive protein-intake guidelines (as much as 2.0 to 3.0 grams per kilogram per day) prescribed by some.

BEST PROTEIN SOURCES

Low-fat dairy products such as skim milk and yogurt, plus grilled chicken and fish and lean red meats provide you with the best protein value for your calorie. For example, a three-ounce piece of lean red meat such as tenderloin contains only 180 calories and twenty-five grams of high-quality, complete protein. A glass of skim milk contains about ten grams of complete protein and almost zero fat. If you prefer not to eat much meat (like me) or dairy products, whey protein powder mixed into skim milk, 100 percent pure fruit juice, or water is an excellent source of high-quality protein. Incomplete proteins—sources that do not contain all twenty amino acids—are also useful when eaten in combination. This is of particular importance to

Table 11.1 Caloric Content of Macronutrients

Macronutrient	Calories/gram
Carbohydrate	4
Protein	4
Fat	9

vegetarian athletes, who, by the way, are more likely to be protein-deficient.

Fat

It's true that most Americans eat far too much fat, which contributes to our high incidence of heart disease, cancer, hypertension, and obesity. Still, getting too little fat has serious implications as well. Dietary fat is necessary as a source of essential fatty acids, which are involved in critical physiological processes such as the functions of the immune system and hormone production. Furthermore, our cell membranes consist largely of phospholipids (fatty acid derivatives), without which we would not be able to make healthy new cells, including muscle cells. A dietary fat deficiency in female athletes has been shown to cause amenorrhea (menstrual cycle irregularities), which may affect the development and maintenance of bone tissue.

DAILY REQUIREMENTS

On average the body's minimum fat requirement is fifteen to twenty-five grams per day. Usually fat-intake recommendations are expressed in terms of percentage of total calories consumed daily. For climbers, 15 to 30 percent of total calories should come from fat, depending on your climbing preference.

For cragging and bouldering, where a low percentage of body fat is desirable and the energy demands are largely anaerobic, fat intake should be restricted to 15 to 20 percent of total calories

consumed. Alpine climbers, however, may be better off consuming up to 30 percent of daily calories from fat. These endurance climbers place great demands on the larger muscles of the body (especially the legs) and expend much more energy per day than, say, sport climbers. Fat is more calorie-dense than carbohydrate and protein (see table 11.1), and it's a good fuel for long, slow aerobic activities. Both these attributes make foods with a higher fat content more advantageous for alpine climbers than for crag climbers.

FOUR TYPES OF FATS

In consuming your daily requirement of fat, it's important to know which of the four types of fat—saturated, monounsaturated, polyunsaturated, and trans fatty acids—are “good” and “bad.” Although each contains the same nine calories per gram consumed, they are not all created equal in terms of their role in performance nutrition. Consequently, it's important not only to eat the optimal amount of fat but also to have the best ratio among the different types of fatty acids.

- **Saturated fats** are most common in animal products such as milk and dairy, meats, and poultry. They are also present in significant amounts in some nuts, including Brazil and macadamia. Although excessive saturated fat intake does increase serum cholesterol, in particular the LDL or “bad” cholesterol, a certain amount is needed by our bodies to be made into fatty-acid-containing compounds such as hormones and phospholipids.
- **Monounsaturated fatty acids** are found in vegetables and oils including canola, olive, peanut, and avocado. These “monos” are thought to be the most beneficial in protecting against heart disease because of their ability to lower LDL without reducing HDL (the “good” cholesterol).
- **Polyunsaturated fatty acids** are common in fish, especially tuna, mackerel, salmon, and trout, and in corn, sunflower, and soybean oils.

The omega-3 “polys” found mainly in fish and flaxseed are currently being investigated for their roles in fighting inflammatory diseases such as arthritis and other illnesses, including migraine headaches and heart disease.

- **Trans fatty acids** are found in trace amounts in almost all sources of natural fats, but most of those in our diet come from hydrogenated oils. During the process of hydrogenation, liquid vegetable oils are converted into solids by bombardment with hydrogen atoms (as in the making of margarine and shortening). Hydrogenation in effect converts unsaturated fatty acids into saturated fatty acids largely through the formation of trans bonds. Recent studies have raised concerns about these bonds because they produce effects similar to those of saturated fats and may, in fact, be cancer causing.

While most well-trained athletes have a very healthy cholesterol profile (unless they smoke or have a genetic predisposition to problems), it's wise to limit your intake of trans fatty acids. Unfortunately, hydrogenated oils and partially hydrogenated oils are found in such a wide range of foods that they are hard to avoid. For instance, many of the breads, cookies, and snack foods you buy off the shelf at the grocery store contain high amounts of these harmful oils. (Read the labels—you'll be surprised how many foods you eat daily contain them!) Many commercial fried foods are cooked in oils that contain trans fatty acids. Surprisingly, even some health foods and energy bars contain partially hydrogenated oils.

In terms of fat consumption, the best approach is to consume about equal amounts of saturated, monounsaturated, and polyunsaturated fatty acids and minimize intake of trans fatty acids. Try to eliminate hydrogenated oils from your diet, and in doing so you will reduce your consumption of these unhealthy fats to acceptable levels. (Unfortunately, it's very difficult to escape them completely in this world of highly processed and prepackaged foods.)

Examine the labels of the foods you eat most regularly and determine which items are doing you the most nutritional damage. Chances are you can make huge strides by eliminating just a handful of killer items like french fries, fried meats, snack cakes and muffins, salty snacks (buy baked chips and crackers instead), and any highly processed food designed for maximum shelf life (no doubt high in hydrogenated oils).

Carbohydrate

Although fat and protein can be used to provide energy, carbohydrate is the most efficient and effective source of energy for the muscles and brain. A high-carbohydrate diet is also important for athletes due to its protein-sparing effect. If you do not consume enough carbohydrates to meet your energy needs, muscle protein will be broken down for energy—the last thing any strength or power athlete would ever want! Consequently, the popular low-carbohydrate diets (Atkins, Zone, Ketogenic, and so forth) are inappropriate for most active climbers.

Carbohydrates come in two forms: sugars and starches. The sugar foods include fruit, sugar, soda pop, jam, honey, and molasses, while common starches are breads, rice, and pasta. Because these are the best sources of energy for high-intensity training and climbing, I'm sure you are already consuming plentiful amounts of these foods. Not all carbohydrates are created equal, however: Different carbohydrates release sugar into the bloodstream at different rates. The very best athletes know how to leverage this information to maintain stable energy throughout the day and to significantly increase the rate of recovery after a hard workout or day of climbing. If you are serious about climbing better, this subject should be of great interest to you. Enter the glycemic index.

GLYCEMIC INDEX

Historically, nutritionists classified carbohydrates only into the two basic groups of simple carbohydrates (sugars) and complex carbohydrates (starches).

Simple sugars were said to produce a rapid rise in blood sugar and quick energy, while complex carbohydrates were said to provide slow, steady energy. Although this concept holds true in general, recent studies have found that there is a large variability in the rise in blood sugar following the ingestion of various foods from both the sugar and starch groups.

To investigate and more accurately classify the metabolism of carbohydrates, researchers developed the glycemic index (GI)—a powerful nutritional tool I first introduced to climbers in my 1994 book *Flash Training*. This index determines how the ingestion of a particular food affects blood sugar levels in comparison with the ingestion of straight glucose. Consumption of high-GI foods causes a rapid increase in blood sugar and a large insulin response. Low-GI foods produce more subtle changes. Climbers can use knowledge of the glycemic index to control energy levels and to speed recovery after a workout. Here's how.

Stable insulin levels are optimal for long-duration stop-and-go activities such as all-day climbing and long training sessions. Experts also agree that a steady insulin curve promotes muscle growth and discourages fat storage. This makes low- to medium-GI foods preferable for climbers in most situations. High-GI foods produce large swings in blood sugar and an insulin spike. One minute you are jonesing to crank another hard route and the next you're yawning and feeling strangely weak.

Figuring the glycemic index of certain foods is more difficult than it might seem at first. For instance, most foods classified as simple carbohydrates (cereal, candy, some fruit juices) are high-GI foods. However, so are potatoes, white rice, bread, and bagels—all considered complex carbohydrates. Low-GI foods include vegetables, whole grains, brown rice, and milk (see table 11.2).

As a general rule, the more processed and easily digestible a food, the higher its glycemic index (for instance, liquids have a higher index than similar food in solid form). High-fiber foods tend to elicit

Table 11.2 Glycemic Index of Common Foods

High (> 70)		Medium (50–70)		Low (< 50)	
Sports drinks	70–85	Banana	55	Balance Bar	30
Clif Bar	70+	PowerBar	65	Peanuts	14
Bagel	72	Raisins	64	Apple	38
Carrots	71	Granola bars	61	Orange	43
Corn chips	73	Macaroni	64	Pear	36
Cornflakes	77	Shredded wheat	58	Grapefruit	23
Doughnut	76	Sweet potato	54	Yogurt (w/ fruit)	30
Honey	73	Bran muffin	60	All-Bran	42
Jelly beans	80	Oatmeal	61	Whole wheat	37
Potatoes	83	Wheat crackers	67	Spaghetti	41
Rice (instant)	91	Cookies	60	Beans	48
Rice cakes	82	Orange juice	57	Lentils	28
Cracker (soda/water)	76	Soft drinks	68	Milk (skim)	32
Glucose	100	Sucrose	65	Fructose	23

a slow insulin response and have a relatively low GI. Finally, foods containing some protein and fat along with carbohydrates come in lower on the scale.

This last piece of information is useful if you don't have the gumption to memorize and use this index. Consuming some protein and fat during each of your carbohydrate feedings serves to moderate the overall glycemic response of the meal. So for a long day at the crags, select a sports drink that contains some protein and pack mainly balanced-type energy bars (those containing nearly equal amounts of calories from protein, carbohydrate, and fat).

The one good time to consume high-GI foods such as juice, soda pop, and most sports drinks (like Gatorade) is at the end of your workout or day of climbing. Intense exercise primes the muscles to immediately reload energy reserves in the form of glycogen. High blood sugar and the insulin spike help drive this repletion process. The optimal window for these high-GI foods is the first two hours following exercise. After that, favor low- to

medium-index foods for slow, steady refueling. See chapter 12 for more information on this and other recovery strategies.

DAILY REQUIREMENTS

Carbohydrates should account for nearly two-thirds of your daily calorie intake. This means that roughly two-thirds of your plate should be covered with pasta, rice, potatoes, and vegetables, with the other third comprising lean, protein-rich foods. Be sure to apply the same rules when snacking. Try to pair up carbos such as a bagel or fruit with some protein like skim milk, yogurt, or a small amount of peanut butter. The protein helps slow down the digestion of carbohydrates and results in longer-lasting energy.

You can also calculate your approximate need for carbohydrates according to your body weight. Training for two hours per day, you would need roughly seven grams of carbohydrate per kilogram of body weight. For example, if you weigh seventy-two kilograms (160 pounds), the requirement would be for

Table 11.3 Estimated Nutrient and Calorie Needs

Climber	Macronutrient	Grams Needed	Calories	Total
Male (160 lbs./72kg), active day	carbohydrate	520g	2,080	3,170
	protein	115g	460	
	fat	70g	630	
Male (160 lbs./72kg), rest day	carbohydrate	360g	1,140	2,230
	protein	85g	340	
	fat	50g	450	
Female (110 lbs./50kg), active day	carbohydrate	350g	1,400	2,170
	protein	80g	320	
	fat	50g	450	
Female (110 lbs./50kg), rest day	carbohydrate	250g	1,000	1,582
	protein	60g	240	
	fat	38g	342	

approximately 500 grams of carbohydrates; at four calories per gram of carbohydrate, this would equal 2,000 calories. Engaging in a full day of strenuous climbing, however—compared with two hours of training—may demand as much as ten to fourteen grams per kilogram of body weight.

Water

Water may be the most important nutrient to get right when you are climbing, yet I sense that many climbers are chronically dehydrated. Dr. Kristine Clark, director of sports nutrition at Penn State's Center for Sports Medicine, says that "even a 1 to 2 percent drop in water will cause problems in performance." The earliest symptoms of mild dehydration are a loss of concentration and enhanced fatigue. Clark adds that "a 3 percent drop in water level can create headaches, cramping, dizziness." Furthermore, a recent study found that dehydration leading to just a 1.5 percent drop in body weight resulted in a statistically significant drop in maximum strength (Schoffstall 2001).

In a sport as stressful as climbing, dehydration also increases your chance of a joint or tendon injury. Consider that proper hydration facilitates transport of nutrients to the cells, helps protect tissues from injury, and maintains joint lubrication. Therefore, for the purpose of injury prevention, maintaining proper hydration as you train or climb is as important as a proper warm-up.

RECOMMENDATIONS FOR PREVENTING DEHYDRATION

As a rule of thumb, it's a good idea to prehydrate ("camel up," as it's called) before you go to the gym or head out climbing by drinking two cups (about sixteen ounces) of water. Follow this with a minimum of an eight-ounce serving every hour or two throughout the day. This would total one to two quarts of water consumed over an eight-hour period on the rocks. This is a bare-minimum amount—what you might drink, say, on a cold day when you perspire very little. Climbing on a humid, eighty-degree day, however, would roughly double

this requirement. That means carrying three to four quarts of water with you for a hot eight-hour day of climbing. Of course, I doubt you know of very many, if any, climbers who carry this amount of liquid to the crags—thus the mass of climbers are unknowingly detracting from their climbing ability and increasing injury risk due to mild dehydration.

Optimal Macronutrient Ratio

As you may have gleaned from the previous sections, the optimal macronutrient ratio for a climber depends on the type of climbing activity. High-intensity, stop-and-go climbing like bouldering or cragging is best fueled with a 65:15:20 ratio of carbohydrate, protein, and fat, respectively. The long, slow distance training and climbing of an alpine climber demands a higher total calorie count per day than the typical rock climber, and this need would be more easily met with a higher-fat diet (though adequate carbohydrates are still necessary). Consequently, a macronutrient ratio of 55:15:30 would be more suitable for the alpine climber. See

Diet and Nutritional Tips

1. Strive for a 65:15:20 caloric ratio of carbohydrate, protein, and fat in your overall daily diet.
2. Consume 1.2 to 1.5 grams of protein per kilogram of body weight per day to facilitate maximum recovery and muscular regeneration.
3. Limit fat consumption to mono- and polyunsaturated fats from vegetable and fish oils. Avoid saturated and trans fats, fried foods, and pre-packaged snack products.
4. Consume several small carbohydrate snacks throughout the day to provide sustained energy, positive mood, and clear thinking. Favor low- to mid-GI foods.
5. Drink an eight-ounce serving of water every hour or two throughout the day—and twice this amount on hot days.

table 11.3 for sample calorie counts for a typical male and female rock climber, but realize that you personally could have significantly different requirements.

Micronutrients and Sports Supplements

The subject of micronutrients and sports supplements is so broad that it's impossible to discuss in a comprehensive manner in an instructional text like this. Still, the use of vitamin supplements and functional foods is so common in sports that the subject is worthy of at least a primer.

Sales of vitamin and sports supplements is a multibillion-dollar industry that bombards us with multilateral, never-ending advertising. Some of the weight-loss and strength-gain claims are truly remarkable, but very few of these claims are backed by reliable scientific studies. Many products are promoted with strong anecdotal claims and well-paid pitchmen, but very few actually do what the advertisers claim.

It's my belief that up to 90 percent of the sports supplements on the market are nothing other than modern-day snake oil, yet a serious athlete looking to maximize performance must be careful not to throw the baby out with the bathwater. On the pages that follow, we'll sort out the handful of products that could potentially enhance your training response and climbing performance.

Micronutrients

Vitamins and minerals are the essential dietary micronutrients. Although the body needs only very small amounts of these nutrients (compared with the macronutrients), they do play a vital role in almost every bodily function—from muscular growth and energy metabolism to neural conduction and memory. Consequently, your health and athletic performance can suffer in a number of ways if you are not consuming enough of these micronutrients.

Studies have shown that as much as two-thirds of self-selected diets contain less than the recommended daily allowances (RDA) of certain vital vitamins and minerals. Furthermore, despite being updated in recent years, RDAs are still believed by many experts to represent low-end requirements for a serious athlete. Therefore, taking a daily multivitamin would be prudent for any serious athlete. It's a remarkably cheap nutritional investment—a generic multivitamin, purchased at CVS or Walgreens, costs less than \$15 for a three-month supply.

Magnesium and zinc are two important minerals shown to be consumed in less than recommended amounts by a majority of the population. Although both are included in most daily multivitamin brands, they aren't well absorbed, and a deficiency in these minerals could mean you're getting shortchanged on training response. Several recent studies have shown a statistically significant increase in muscular strength in a group of athletes taking supplemental magnesium and zinc (in the form of a patented supplement known as ZMA) versus a control group of athletes taking a placebo (Brilla 1998). Clearly, any climber engaging in serious strength training could benefit from taking ZMA or by consuming foods rich in magnesium (spinach, nuts, beans and lentils, whole grains, and low-fat dairy) and zinc (seafood, beef, cashews, pork, and chicken).

Calcium and iron are two other minerals that some climbers may be lacking. Vegetarians often fall short on iron consumption, and some women may not get enough iron and calcium in their diets. A multivitamin is the best way to obtain extra iron, especially since its absorption is enhanced in the presence of vitamin C (also in a multivitamin). Females wanting to get extra calcium in their diets could take a calcium supplement (Tums are cheap and taste good!) or simply drink a few glasses of skim milk each day.

Sports Supplements

The most effective way to increase climbing performance is through a long-term, dedicated effort to

improve your technique, mental control, and upper-body strength. Unless you are working to dial in each of these areas, the few sports supplements that do work are probably a waste of money. If you are actively honing your technical skills and training to increase your physical and mental fitness, however, you may be able to further improve your performance through the use of a handful of ergogenic (performance-enhancing) supplements.

PROTEIN POWDERS

As discussed earlier, athletes have a greater daily protein requirement than sedentary people, since strenuous exercise results in a higher protein turnover. While mega amounts are not necessary (as some supplement companies and fitness magazines suggest), a 160-pound climber does need to consume between 86 and 108 grams of protein per day. This modest amount can be met through a well-rounded diet; it may be tough to consume adequate protein, however, if you do not eat meat such as chicken, fish, and lean red meats. There is also the limiting factor relating to the biological value of the protein source consumed. Not all protein sources are equal when it comes to providing your body with the necessary building blocks.

Consequently, scientists have developed a number of ways to measure the quality of protein sources. The biological value (BV) is one of the most commonly used; it's based on how much of the protein consumed is actually absorbed and utilized by the body. The higher the BV, the greater the amount of protein that is actually available for use by the body to strengthen muscles and connective tissues, and to support enzyme formation, among other things.

When the BV scale was originally developed, the egg was considered the perfect protein source (remember Rocky?) and sat alone atop the chart with a value of one hundred. Since then new technologies have enabled the creation of super-proteins that are equally valuable to the body but without the fat found in many high-protein foods. Whey

Table 11.4 Biological Values of Protein Sources

Food	BV
Whey	104
Whole egg	100
Milk	91
Egg white	88
Fish	83
Beef	80
Chicken	79
Soy	74
Beans	49
Peanuts	43

Not all protein sources are equal when it comes to providing your body with the necessary building blocks. Select mostly high-BV foods from the top half of the list.

is the current superstar of proteins—its BV is 104, meaning that it provides even better nitrogen retention than eggs. Therefore, whey protein could be viewed as superior to the lower-BV foods such as fish, beef, chicken, or soy in helping meet the protein needs of an athlete.

While whey protein costs approximately \$0.60 to \$0.90 per twenty-gram serving, the investment is a good one considering its ease of use and high BV. Many brands are available from health food stores, supplement catalogs, and online merchants, so shop around for the best deal—look specifically for a blend of “whey isolate and whey concentrate,” which is the highest quality and mixes easily in milk, juice, or water. Such liquid protein is ideal first thing in the morning, and it can increase your rate of recovery when consumed immediately following your workout or a day of climbing (more on this in chapter 12).

One of the highest-quality and most affordable protein sources is skim milk (see table 11.4). Since each glass contains ten grams of protein, consuming

a quart of milk per day would provide forty grams toward your total protein requirement. As someone who only occasionally eats meat, I have relied on skim milk as my primary source of protein for the last thirty-five years (since I first read that John Gill slammed back milk protein after climbing). In drinking about two gallons of milk per week for more than thirty-six years, I’ve somehow consumed nearly 4,000 gallons of skim milk since I started climbing. All I can say is, “Got milk?”

Even those who are lactose-intolerant can consume milk without any nasty side effects by purchasing acidophilus milk. Regardless, be sure to always select 1 percent or skim milk (best), since whole milk contains a significant amount of saturated fat.

SPORTS DRINKS

Since the invention of Gatorade in the early 1970s, sports drinks have grown into a massive industry. It’s now hard to find a serious athlete who does not consume sports drinks to help replenish energy. Dozens of different sports drinks are available in bottled form, including Gatorade and Powerade, the two most popular drinks on the market right now. Many others are sold in a bulk powdered form ideal for mixing up at home before a workout or day at the crags.

As you might expect, not all sports drinks are the same. Some are merely glorified sugar water, while others include electrolytes or any number of vitamins, minerals, herbs, or other nutrients said to increase athletic performance. Clearly, many of the claims are unsubstantiated—and at up to three bucks a bottle, some are just a big rip-off.

The active ingredients in these products fall into two main categories: electrolytes and fuel replacements (although some drinks like Red Bull are laced with caffeine, herbs, and vitamins, too). A simple understanding of both will help you understand how these drinks might be of benefit to you.

Electrolytes such as potassium, magnesium, calcium, sodium, and chloride are critical for



Snack and rehydrate every hour or two. You'll climb better all day long and top out on long routes with energy to spare!

concentration, energy production, nerve transmission, and muscle contraction. Fortunately, electrolyte loss during exercise is quite slow, so even a full day of climbing won't cause significant depletion. A reasonable diet and multivitamin provide you with all the electrolytes you need for ordinary training or a day of climbing. Still, a sports drink with electrolytes may be beneficial if your food supply will be limited for a few days, as in big-wall or alpine climbing.

Fuel sources in the sports drinks are mainly carbohydrates, including glucose, sucrose, fructose, maltodextrin, and lactates. Glucose and sucrose (table sugar) are the fuel sources in the original

sports drink Gatorade and have since been adopted by many other companies. Ironically, some athletes now shun drinks with large amounts of glucose and sucrose to avoid a blood sugar spike. Unless you are engaged in an activity that continues steadily for a couple of hours (such as long-distance biking, running, or hiking), this quick increase in blood sugar may be followed by an energy crash as insulin kicks in to reduce the elevated blood sugar level. In a stop-and-go sport like climbing, this pullback on blood sugar can leave you feeling more "flat" and lethargic than before you consumed the drink!

Consequently, you should avoid any drink whose first ingredient (after water) is glucose or high-fructose corn syrup—both are signs that the drink has a high glycemic index and will release sugar rapidly into the blood. Instead, shop around for powdered sports drink mixes that list fructose (not the same as high-fructose corn syrup) or a protein source such as whey as the first or second ingredient. These have a lower glycemic index and hence provide a slower, more sustained release of carbohydrate ideal for climbers. Accelerade by Pacific Health Labs or PowerBar's Performance Recovery are two good choices.

Carbohydrate (glycogen) depletion in the muscles and liver is a primary cause of fatigue when performing long-duration activities of more than ninety minutes. The traditional use of sports drinks is for situations when additional fuel is needed for prolonged activity lasting more than ninety minutes. Therefore, consuming a fructose-based or protein-containing sports drink during the course of a day at the crags will help maintain energy levels throughout the day, though consuming some food would be beneficial as well. Conversely, an hour or two of bouldering or a short gym workout will not benefit from the added fuel of a sports drink.

While a slow-release, fructose-based or protein-containing sports drink is ideal for when you are climbing, it might be better to consume a high-GI drink upon completion of your workout or day of climbing. Numerous studies have shown that

elevating blood sugar as soon as possible after exercise provides the substrate for synthesis of muscle glycogen (Robergs 1991). This is especially important if you plan to climb the very next day—replenishing glycogen stores takes up to twenty-four hours, so you need to get the process started immediately and at high speed (see chapter 12 for more recovery tips).

ENERGY BARS

Energy bars have been a dietary mainstay of many climbers since PowerBar hit the scene back in the late 1980s. Originally intended for endurance athletes like bikers and runners, many of the energy bars are designed to deliver a rapid release of sugar into the bloodstream. Therefore, many energy bars possess a high glycemic index (greater than seventy); these are easily identified by their first ingredient, high-fructose corn syrup. Climbers in search of sustained energy are better off sticking to foods with a GI of less than seventy, since they provide more sustained energy and less of an insulin response.

The numerous balanced-style bars that have entered the market in recent years typically possess a glycemic index in the forty-to-sixty range (my favorites are Balance Bar and PowerBar Protein Plus). The higher amount of protein and fat contained in these bars helps slow the release of sugar into the bloodstream. The balanced 40:30:30 macronutrient ratio also helps conserve glycogen and may even help spare muscle protein from being used for energy during a long day of climbing. Consequently, consuming a couple of Protein Plus PowerBars (Fig Newtons are also good) and drinking lots of water may be the single best combination to maintain energy, spare muscle protein, and prevent dehydration.

CREATINE

There are dozens of sports supplements that claim to increase strength/power and to help build muscle. While most are, in fact, worthless, creatine has been shown to produce increases in muscular

strength in dozens of well-executed studies. Based on this, creatine must be a good supplement for climbers. Right? Not so fast, my highball-sending, redpoint-crushing friend!

Let's look at the facts. First, creatine is by far the most effective, legal sports supplement on the market. Not only has it been shown to enhance explosive strength in numerous well-controlled studies (Toler 1997 and Kreider 1998), but when consumed in large doses the user actually increases lean mass (large, harder muscles). Consequently, creatine has become the biggest selling sports supplement in the country, and it's widely used by football and baseball players, weight lifters, bodybuilders, MMAers, Cross-Fitters, sprinters, and millions of fitness buffs.

Creatine is a compound that's natural to our body and, as you learned in chapter 5, creatine phosphate it is used in the muscles to help create ATP (the energy source for brief, explosive movements) as part of the anaerobic alactic energy pathway. Creatine is present in animal foods such as red meat; however, the amount consumed in a normal diet is quite small (typically just a gram or two per day). Studies have shown that taking twenty grams per day of supplemental creatine for five or six days will enhance performance in short-duration, high-intensity exercise such as sprinting or weight lifting. This "creatine loading" protocol is the method used by most athletes in search of greater strength/power and gains in lean body mass. (Long-term use of the creatine loading protocol may be stressful on the kidneys, so it would be wise to cycle on and off creatine if you are consuming "loading" amounts.)

Two side effects of creatine loading are weight gain and "cell volumizing." Both these effects occur because creatine associates with water as it is stored in the muscles (95 percent of the creatine is stored inside muscle cells, and so intracellular water almost certainly increases). Over the six-day loading phase, more and more creatine is stored, and an increasing amount of water is drawn into the muscle cell—this gives muscles a fuller, "pumped" feel and look, just what bodybuilders and fitness buffs want. This

loading process, therefore, typically results in a water weight gain of two or three pounds (or more). This is a good thing for athletes in sports where increased lean body mass and speed (inertia) can be used to your advantage (e.g., football, swinging a bat, or swinging your fist). However, in a sport that requires a high strength-to-weight ratio, it can have a negative impact on performance.

Some climbers have argued that stronger muscles (due to creatine loading) can easily lift the extra weight gained. The problem, however, is that creatine loads in all muscles of the body, not just the “climbing” muscles. Furthermore, creatine will load proportionally more in the largest muscles of the body—the legs!—not exactly the place smart climbers would want to increase muscle mass. Therefore, I’m sure you’ll agree that the creatine loading protocol would likely be counterproductive for a serious climber.

If you are still not convinced, let’s consider the cell volumizing effect of heavy creatine use. Bodybuilders love the fact that their muscles pump up more easily when they are loaded with creatine (and water). I quickly noticed this same effect when experimenting with creatine when it first appeared on the market in 1993. It seemed strange at the time, but I pumped out more quickly when I was “on” creatine, despite the fact that I felt like I had a little more zip in my muscles. What I theorize is that the cell volumizing effect may lead to faster occlusion of important capillary blood flow within the muscles during strenuous exercises, thus causing a severe pump and rapid muscle failure when the energy pathway shifts away from anaerobic alactic (after ten to fifteen seconds). So while creatine use may have a positive effect on brief powerful climbs lasting less than fifteen seconds, the “pump effect” may very well hamper climbing performance on longer boulder problems and difficult roped ascents.

All this said, I do believe that well-timed, small doses of creatine can help climbers (especially boulderers) slightly increase their anaerobic power and

The Truth About Creatine for Climbers

Small-dose creatine usage (two to five grams per day) may yield a slight increase in power and a small edge in recovery between strenuous exercises, boulder problems, and routes, without any significant undesirable side effects.

The high-dose “creatine loading” protocol (ten to twenty grams consumed daily) is likely counterproductive for climbers, due to weight gain (commonly two to four pounds) and increased muscle “pump” during long boulder problems or routes.

A small percentage of individuals are creatine “nonresponders,” meaning that creatine consumption results in no observable positive or negative effects (possibly because they have an unusually low percentage of fast-twitch muscle fibers).

Creatine usage can cause cramping, particularly in the foot and leg muscles, in some individuals. Long-term creatine use may be harmful to the kidneys.

Creatine is not a banned substance, nor is it on the World Anti-Doping Agency (WADA) list of prohibited substances.

recover a bit more quickly. The protocol I’ve developed and used for several years now is to add just two to five grams of creatine to a quart of sports drink that I sip throughout my workout (or climbing day). This provides just a few grams more than a normal “dietary dose” of creatine—enough to hasten recovery of maximal power between exercises and boulder problems, yet likely without the undesirable side effects of “loading.”

CAFFEINE

If you are a coffee aficionado like me, then you’ll surely enjoy this section. Your addiction—er, culinary ritual—may actually give you an extra edge both in training and climbing!



Moderate caffeine consumption can energize your training and, perhaps, enhance performance slightly. Here, Matilda Söderlund is amped to be training at Café Kraft! HANNES HUCH

Caffeine (the active ingredient in coffee) is the number one ergogenic aid of the common man and athletes alike. It's caffeine's well-documented effects on the central nervous system that help you wake up in the morning, gather your focus at school or work, and become a bit more powerful at the gym. Remarkably, caffeine has also been shown to have numerous positive impacts on health—most notable is its protective effect when it comes to neurodegenerative diseases (Parkinson's and Alzheimer's) and cancer (bladder, pancreatic, breast, ovarian, colon, and colorectal).

If you consume a daily coffee, espresso, tea, or another caffeine-laced drink like Red Bull or Mountain Dew, then you know firsthand the stimulant effects of caffeine. But there are several other ways that caffeine might help you train and climb harder, as well as a few undesirable side effects that you need to be aware of. Let's sort things out.

Here are the most notable performance-enhancing effects of caffeine, as summarized in an excellent meta-analysis research paper by Bülent Sökmen, et al. (2008).

- Improved concentration, alertness, and mood, especially in sleep-deprived athletes.
- Enhanced reaction time and improved accuracy in skilled movements.
- Lowered peripheral pain perception and rating of perceived exertion.
- Increased work output and time to exhaustion during aerobic-powered endurance exercise (e.g., endurance routes and multipitch climbs).
- Increased power output and isokinetic strength in maximum-strength and power events lasting less than ten seconds (e.g., short, maximal boulder problem, campus training, one-arm pull-ups).

Table 11.5 Caffeine Content of Common Energy Drinks

Product	Serving Size	Caffeine (mg)
Cola (Coke, Pepsi, Ale 8-1)	12 oz.	approx. 35
Tea	8 oz.	25–50
Mountain Dew	12 oz.	54
Red Bull	8.5 oz.	80
Monster Energy Drink	16 oz.	160
Rockstar Energy Drink	16 oz.	160
Coffee “weak” (McDonalds)	16 oz.	160
5-Hour Energy Drink	2 oz.	200
Anhydrous caffeine (No-Doze, Vivarin)	1 tablet	200
Coffee “strong” (Starbucks, Caribou)	16 oz.	300

Considering these cognitive and physiological effects, it’s not surprising that so many climbers consume caffeine, not only in the morning but also upon arriving at the gym or crag. Still, you must be careful not to develop a more-is-better mind-set when it comes to quaffing your favorite caffeinated drink—there are some potential negative impacts to consider.

- High doses can make you jittery and nervous—not ideal for a delicate face climb (requiring fine motor control) or before competition.
- High doses may impart a generalized ill feeling due to gastrointestinal distress, headache, and perhaps even heart palpitations.
- Given caffeine’s half-life of four to six hours, consumption in the afternoon or evening may lead to insomnia or a restless night’s sleep.
- High doses of caffeine have been shown to lower blood pH (more acidic), and therefore it may negatively impact high-intensity exercise performance relying on the glycolytic energy system (e.g., short, pumpy sport routes).

Vitamin and Sports Supplement Tips

1. Take a daily multivitamin each morning with breakfast.
2. Consider consuming extra protein (especially if you eat little meat or dairy products) by drinking a whey protein shake each morning and evening.
3. When climbing, snack on fruits, balanced-style energy bars, sports drinks (choose those possessing some fructose and/or protein), and other lower-GI foods. Consider adding two to five grams of creatine (never more) to your bottle of sports drink to further enhance recovery.
4. Small, well-timed doses of caffeine and creatine may provide an edge in both training and climbing.
5. Avoid mass-marketed sports supplements, especially those with the wildest claims—most are frauds.

So how much caffeine should you consume to gain a performance edge? Ultimately that depends on your tolerance to this drug, which is a function of the amount you habitually consume daily. Across the research literature, subjects consumed between 3 and 13 milligrams of caffeine per kilogram of body weight; however, low to moderate dosing of 3 to 9 milligrams per kilogram seems optimal (most upsides with fewest negative side effects). For the average-size climber, this equates to consuming somewhere between 200 and 600 milligrams of caffeine—use the low end of this range if you are caffeine “naïve” and upper-range dosing if you are a regular, heavy consumer. See table 11.5 for approximate caffeine content in common beverages.

The bottom line: Caffeine use can potentially enhance your training and climbing performance, but caffeine “overdose” will likely hurt your performance. The key is to experiment with and understand caffeine’s unique effects on you. Ideally, you’ll learn to manipulate your caffeine dosage to get the best performance-enhancing effect via moderate dosing—that is, decrease daily dosing in the days leading up to a critical climb or competition, then return to your “normal” moderate dose on the day of the event. This is a superior approach to simply doubling your caffeine dose on the performance day, thus risking the negative side effects listed above.

SPORTS SUPPLEMENTS THAT DON'T WORK

The list of sports supplements that don’t work is too long to detail here. I will, however, name a couple of the most popular, most hyped supplements that have little or no reliable research to back up the big claims.

First and foremost is anything marketed as a “testosterone booster.” They don’t work—period. Fortunately, you can naturally increase your testosterone levels by eating a protein-rich diet, exercising intensely, shunning alcohol and sweets, and getting enough sleep. Next on the list of “bogus supplements” are nitric oxide boosters, most of which yield no effect at all; the exceptions are citrulline malate and beet root concentrate, both of which may support nitric oxide production and produce a slight beneficial vasodilation effect.

Let me cut to the chase: Pretty much everything touted as a “performance enhancer” or “miracle cure” is likely nothing more than expensive snake oil. Certainly there are performance-enhancing drugs that could help you train harder and perform better, but why would you want to go down that path? Outside of caffeine and creatine—by far the two most effective sports supplements—any substance that truly aids performance in a significant way is either illegal or on the WADA prohibition list.



*Matty Hong on the
first ascent of La
Cucaraicha (5.13+),
Rifle, Colorado.*
MATTY HONG
COLLECTION